

CARHy-WSN: A Context-Aware Hybrid Proactive-Reactive Routing Protocol for Heterogeneous IoT-Driven Wireless Sensor Networks

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Abstract

Wireless Sensor Networks (WSNs) deployed in IoT-driven smart city environments carry heterogeneous application traffic with fundamentally different quality-of-service requirements. Existing hybrid routing protocols switch between proactive and reactive modes based on a single network metric, making them insensitive to application urgency and unable to simultaneously optimise energy consumption and end-to-end latency across different traffic classes. This paper proposes CARHy-WSN, a Context-Aware Hybrid Routing protocol that introduces a three-dimensional context vector — comprising application urgency class, normalised residual energy, and link stability score — to dynamically select the optimal routing mode per packet. CARHy-WSN is implemented in MATLAB and evaluated against five baseline protocols under three traffic classes and a mixed-traffic scenario in a 100-node heterogeneous WSN simulation. Results demonstrate that CARHy-WSN achieves an 82.5% reduction in Class A emergency packet latency compared to AODV, a 70.2% reduction in routing overhead compared to DSDV, and maintains a packet delivery ratio of 100% across all evaluated scenarios, while preserving competitive energy balance throughout all tested configurations. A lightweight decision tree variant achieves 100% classification accuracy with only 0.3 KB memory overhead, enabling self-configuring deployment. CARHy-WSN pri-

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oritises simultaneous latency and overhead optimisation for safety-critical heterogeneous IoT deployments, achieving these gains within a deliberate network lifetime trade-off relative to lifetime-optimised baselines.

Keywords: Wireless sensor networks, Context-aware routing, Hybrid routing protocol, IoT, Smart city, Energy efficiency, QoS, Heterogeneous traffic, MATLAB simulation

1. Introduction

Wireless Sensor Networks (WSNs) have emerged as a foundational component of Internet of Things (IoT)-driven smart city infrastructure, enabling continuous and distributed monitoring across domains including environmental sensing, smart transportation, industrial automation, and emergency response [1, 2]. As smart city deployments scale in both node density and application diversity, the routing protocols governing data delivery face increasing pressure to simultaneously satisfy conflicting requirements: energy efficiency, low latency, reliability, and adaptability to dynamic network conditions.

Existing routing approaches broadly fall into three categories. Proactive protocols, such as DSDV [3] and OLSR [4], maintain routing tables continuously, delivering low latency at the cost of high energy overhead from periodic updates. Reactive protocols, such as AODV [5] and DSR [6], discover routes on demand, conserving energy in sparse networks but introducing significant route discovery delays that are intolerable for time-critical applications. Hybrid protocols, such as ZRP [7], partition the network into zones and apply different modes within and across zone boundaries. However, ZRP and its variants rely on a fixed zone radius parameter and remain insensitive to the nature of the traffic being routed.

In a realistic smart city WSN, a single node may simultaneously generate emergency alerts requiring sub-100 ms delivery, periodic telemetry with a 30-second tolerance, and event-driven uploads with best-effort requirements. No existing protocol adapts its routing behaviour to the urgency class of individual packets. This application-blindness leads to systematic inefficiency: proactive routing wastes energy on low-urgency telemetry, while reactive routing introduces dangerous delays on emergency traffic.

A 2025 comprehensive survey of context-aware routing in IoT-driven WSNs [8] explicitly identifies dynamic application-context-based hybrid mode switching as an open research direction, noting that reactive and hybrid

context-aware approaches have received significantly less attention than proactive ones. No published protocol uses a joint multi-dimensional context vector — comprising application urgency, residual energy, and link stability simultaneously — to trigger full routing paradigm switching. This paper addresses that gap directly.

The main contributions of this paper are as follows:

- A three-dimensional context vector $\mathbf{C} = (U, E_r, L_s)$ comprising application urgency class U , normalised residual energy E_r , and link stability score L_s , and a mode-switching decision table that maps each context combination to the optimal routing mode.
- CARHy-WSN, the first WSN hybrid routing protocol that performs per-packet routing mode selection based on a joint three-dimensional context vector, requiring under 0.1 ms classification latency and under 2 KB RAM overhead per node.
- Evaluation under three simultaneous heterogeneous IoT traffic classes, providing the first routing comparison under realistic smart city mixed-traffic conditions.
- Validation that a lightweight decision tree classifier (depth 3, 0.3 KB memory, 7 comparisons per packet) learns the context-switching rules with 100% test accuracy and 99.96% cross-validation accuracy, demonstrating the feasibility of a self-configuring deployment that requires no manual threshold tuning.

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 presents the system model. Section 4 describes the CARHy-WSN protocol design. Section 5 presents the simulation setup. Section 6 presents and analyses the results. Section 7 discusses key findings and limitations. Section 8 concludes the paper.

2. Related Work

2.1. Proactive and Reactive Routing Protocols

Routing protocols for WSNs and ad hoc networks have been broadly classified into proactive and reactive categories, each with distinct performance trade-offs that make neither approach universally optimal for heterogeneous IoT deployments.

Routing protocols for WSNs have been extensively surveyed and classified [9]. Proactive protocols maintain routing tables continuously through periodic broadcasts. DSDV [3] established the foundational destination-sequenced distance-vector approach, guaranteeing loop-free routes at all times. OLSR [4] extended this paradigm through multipoint relay (MPR) selection, substantially reducing flooding overhead in dense networks. While proactive protocols deliver consistently low latency by eliminating route discovery delays, their continuous maintenance overhead leads to significant energy waste on low-traffic nodes [9].

Reactive protocols discover routes exclusively on demand. AODV [5] pioneered this approach by initiating route requests only when data must be transmitted, conserving energy in sparse or low-traffic conditions. However, the route discovery delay of 20–100 ms introduced by RREQ/RREP flooding renders reactive protocols unsuitable for latency-sensitive applications [10]. A comparative analysis by Nurul Hidayah et al. [11] confirmed that neither proactive nor reactive protocols alone achieve optimal performance across varying network conditions, motivating the need for adaptive hybrid approaches. Recent work [12] demonstrated that AODV, DSDV, and DSR all exhibit significant performance degradation under heterogeneous traffic loads, highlighting the fundamental limitation of single-mode routing for IoT-driven smart city deployments.

2.2. Hybrid Routing Protocols

Hybrid protocols attempt to combine the latency advantages of proactive routing with the energy efficiency of reactive routing. ZRP [7] pioneered zone-based hybrid routing by maintaining proactive routes within a fixed-radius zone and using reactive inter-zone routing for destinations beyond the zone boundary. While ZRP reduces control overhead compared to purely proactive protocols, its fixed zone radius is configured statically and does not adapt to node energy state or traffic urgency — a fundamental limitation that CARHy-WSN addresses directly.

Subsequent hybrid approaches incorporated additional optimisation dimensions. Chaurasiya et al. [13] proposed a hybrid clustering technique for multilevel heterogeneous IoT WSNs, combining cluster-based organisation with multi-hop routing to reduce energy consumption. Lei [14] proposed a hybrid energy-aware approach combining particle swarm optimisation with fuzzy clustering for IoT routing, demonstrating 52% energy reduction compared to

DEEC, though at the cost of requiring offline training data for the fuzzy membership functions. Srinivasa Babu et al. [15] proposed a neuro-fuzzy hybrid routing scheme for IoT that incorporates context-like features, but similarly requires a trained model that cannot adapt at runtime without retraining. Nadanam et al. [16] recently proposed a quantum-optimised hybrid routing framework achieving 30% energy reduction, though the computational complexity of quantum optimisation makes it unsuitable for resource-constrained sensor hardware. Sudha et al. [17] combined Q-learning with machine learning for cluster head selection and routing in 5G-enabled WSNs, demonstrating improved throughput over LEACH, but without considering per-packet application urgency as a routing criterion.

A common limitation across all these hybrid approaches is that mode selection or routing decisions are made based on a single optimisation objective — typically energy or load — without awareness of the application-level urgency of the traffic being routed.

2.3. Context-Aware Routing in IoT-WSNs

Context-aware computing for IoT was formally defined by Perera et al. [18], who identified five contextual dimensions — identity, location, activity, time, and relations — as fundamental to intelligent IoT decision making. This foundational work motivates the use of contextual information in routing decisions, though the specific application to WSN hybrid routing mode selection had not been addressed until recently.

Context-aware routing has since been applied across several IoT and WSN domains. Monteiro et al. [19] proposed context-aware network selection in heterogeneous wireless networks, demonstrating that context-driven decisions outperform static policies, but their work focused on network selection rather than intra-WSN routing mode switching. Akpakwu et al. [20] proposed a context-aware congestion control scheme for CoAP/UDP IoT networks, applying context sensing to transport-layer decisions rather than routing. Sithik and Kumar [21] proposed a multi-context aware parent selection scheme for the RPL protocol using grey set theory, incorporating multiple contextual parameters into routing metric computation. However, their approach targets RPL parent selection exclusively and does not implement dynamic switching between fundamentally different routing modes (proactive, reactive, and hybrid) based on application urgency class. Unlike Sithik and Kumar [21], who optimise a single routing metric within RPL, CARHy-WSN introduces a three-dimensional context vector that triggers full routing paradigm switching

— the first such mechanism reported in the WSN routing literature. Karagiannis et al. [22] proposed context-aware routing for fog computing systems, reducing latency by 23% and hop count by 73% through context-driven forwarding decisions. Their work operates at the fog layer rather than the WSN sensor layer, and does not model heterogeneous traffic classes with conflicting QoS requirements. Hasan et al. [23] proposed a context-aware trust prediction framework for opportunistic IoT routing using Bayesian inference, achieving 88.9% delivery ratio, though the trust-based context dimension differs fundamentally from the urgency-energy-stability context vector proposed in CARHy-WSN.

Smart city IoT routing has also received significant attention. Hanif et al. [24] proposed opportunistic IoT-WSN integration for smart city deployments. Inga et al. [25] addressed WSN sizing and routing for smart city deployments. Ekler et al. [26] proposed energy-aware IoT routing algorithms specifically for smart city environments. Karunkuzhali et al. [27, 28] proposed QoS-aware routing frameworks for IoT-WSNs in smart cities, demonstrating that multi-class service differentiation improves network performance. However, none of these works implement dynamic routing mode switching based on per-packet application urgency. El-Fouly et al. [29] proposed environment-aware real-time routing for multi-sink WSNs in smart cities, incorporating deadline constraints and link quality metrics. While this work shares the multi-sink topology with our prior work [30] and considers real-time delivery requirements, it does not model heterogeneous application urgency classes or implement context-driven mode switching.

Recent surveys confirm that AI-assisted and context-aware routing are among the most active research frontiers in IoT-WSN networking. Priyadarshi [31] reviewed meta-heuristic and AI-based routing approaches, while Priyadarshi [32] further examined AI-driven routing improvements for energy efficiency and reliability. Saadallah and Alabady [33] identified energy-efficient context-sensitive routing as an open challenge in their comprehensive IoT routing study. Yilmaz et al. [34] highlighted the integration of ML and AI into WSN routing as a key trend for 5G/6G sustainable communication. Daanoune and Baghdad [35] recently proposed a reinforcement learning-based clustering and routing protocol (RLCR) for heterogeneous WSNs, demonstrating the potential of AI-driven approaches for adaptive routing, though without application-context awareness in mode selection.

Table 1 summarises the key distinguishing features of CARHy-WSN compared to the most closely related context-aware routing works.

Table 1: Comparison of CARHy-WSN with Related Context-Aware Routing Works

Protocol	Context dimensions	Mode switching	Traffic classes	Urgency aware	Energy model
Sithik & Kumar [21]	2	RPL parent	1	No	No
Karagiannis et al. [22]	2	Fog forwarding	1	No	No
Hasan et al. [23]	1 (trust)	Route selection	1	No	No
Monteiro et al. [19]	2	Network selection	1	No	No
ZRP [7]	1 (zone)	Hybrid fixed	1	No	No
CARHy-WSN (Ours)	3	Full paradigm	3	Yes	Yes

2.4. Research Gaps

The preceding review reveals four specific gaps that motivate CARHy-WSN. First, Khedr et al. [8] conducted the first comprehensive survey of context-aware routing in IoT-driven WSNs and explicitly identified dynamic application-context-based mode switching as an open research direction, noting that reactive and hybrid context-aware methods have received significantly less attention than proactive approaches. Second, Kumar et al. [36] identified application-context awareness as an emerging research gap through a bibliometric analysis of energy-efficient routing in IoT WSNs spanning 2015–2024. Third, Chu et al. [37], Khanna et al. [38], and Prashanth et al. [39] all noted that routing adaptation in heterogeneous WSNs remains dominated by single-objective optimisation approaches that cannot simultaneously satisfy the conflicting requirements of heterogeneous application traffic classes. Fourth, no published protocol uses a joint three-dimensional context vector comprising application urgency, residual energy ratio, and link stability score to dynamically select between proactive, reactive, and hybrid routing modes on a per-packet basis — the specific contribution of CARHy-WSN introduced in this paper.

3. System Model

3.1. Network Topology

Consider a WSN comprising N heterogeneous sensor nodes deployed randomly over a square area of side length A m according to a uniform distribution. Let $\mathcal{N} = \{1, 2, \dots, N\}$ denote the set of all nodes. A base station (BS) is located at the centre of the deployment area, and a secondary

198 sink node is positioned at the right boundary, consistent with the multi-
 199 sink topology in our prior work [30]. Both the BS and sink are assumed
 200 to have unconstrained energy. Each sensor node $i \in \mathcal{N}$ is characterised by
 201 a fixed position (p_i^x, p_i^y) and a heterogeneous initial energy E_i^0 drawn from
 202 a uniform distribution $E_i^0 \sim \mathcal{U}(E_{min}, E_{max})$, reflecting the reality of diverse
 203 sensor hardware in smart city deployments [1].

204 A node i is considered *alive* if its residual energy $E_i(t) > 0$ at simulation
 205 round t , and *dead* otherwise. Dead nodes neither transmit nor receive. The
 206 network is considered operational as long as at least one node remains alive
 207 and connected to the BS.

208 3.2. Traffic Model

209 Smart city IoT deployments generate heterogeneous application traffic
 210 with fundamentally different QoS requirements. This paper models three
 211 traffic classes simultaneously:

- 212 • **Class A — Emergency traffic:** Generated by safety-critical sensors
 213 such as fire detectors, intrusion alarms, and accident monitors. Modelled
 214 as a Poisson process with arrival rate λ_A . Strict latency requirement:
 215 end-to-end delay must not exceed 100 ms.
- 216 • **Class B — Periodic telemetry:** Generated by environmental sensors
 217 reporting temperature, humidity, air quality, and energy consumption.
 218 Modelled as a deterministic periodic process with one packet per minute.
 219 Tolerates delays up to 30 seconds.
- 220 • **Class C — Event-driven uploads:** Generated by motion detectors,
 221 occupancy sensors, and camera triggers. Modelled as a Poisson process
 222 with arrival rate λ_C . Best-effort delivery with no hard latency constraint.

223 Each packet carries a class tag $U \in \{A, B, C\}$ in its header, which is read
 224 by the context classifier at the transmitting node before routing decisions are
 225 made.

226 3.3. Channel and Energy Model

227 Wireless communication is modelled using the first-order radio energy
 228 model [40], which captures the dominant sources of energy consumption in
 229 sensor nodes: electronics and amplification. The energy consumed by node i
 230 to transmit an L -bit packet to a receiver at distance d is:

$$E_{tx}(L, d) = \begin{cases} L \cdot E_{elec} + L \cdot \varepsilon_{fs} \cdot d^2 & d < d_0 \\ L \cdot E_{elec} + L \cdot \varepsilon_{mp} \cdot d^4 & d \geq d_0 \end{cases} \quad (1)$$

231 where $d_0 = \sqrt{\varepsilon_{fs}/\varepsilon_{mp}} = 87.71$ m is the crossover distance between free-space
 232 and multipath propagation models. The energy consumed to receive an L -bit
 233 packet is:

$$E_{rx}(L) = L \cdot E_{elec} \quad (2)$$

234 The residual energy of node i after a transmission round is:

$$E_i(t+1) = E_i(t) - E_{tx}(L, d) - E_{rx}(L) \quad (3)$$

235 subject to $E_i(t) \geq 0$.

236 3.4. Link Stability Model

237 To capture dynamic link quality variations, CARHy-WSN maintains a per-
 238 node *link stability score* $L_s^i(t) \in [0, 1]$, computed as an Exponential Weighted
 239 Moving Average (EWMA) of the ACK success history over a sliding window
 240 of W most recent transmissions:

$$L_s^i(t) = \sum_{k=0}^{W-1} w_k \cdot a_k^i \quad (4)$$

241 where $a_k^i \in \{0, 1\}$ is the ACK outcome of the k -th most recent transmission
 242 from node i , and $w_k = \alpha(1 - \alpha)^k/Z$ are normalised exponentially decaying
 243 weights with smoothing factor $\alpha = 0.3$ and normalisation constant $Z =$
 244 $\sum_{k=0}^{W-1} (1 - \alpha)^k$.

245 3.5. Performance Metrics

246 **Network lifetime:** The First Node Death round $r_{FND} = \min\{t :$
 247 $\exists i, E_i(t) = 0\}$ and the Half Node Death round $r_{HND} = \min\{t : |\{i :$
 248 $E_i(t) = 0\}| \geq N/2\}$.

249 **Packet Delivery Ratio:**

$$\text{PDR} = \frac{D_{delivered}}{D_{total}} \quad (5)$$

250 **Energy balance (Gini coefficient):**

$$G = \frac{2 \sum_{i=1}^N i \cdot c_i}{N \sum_{i=1}^N c_i} - \frac{N+1}{N} \quad (6)$$

251 where c_i is the energy consumed by node i , sorted in ascending order ($c_1 \leq$
252 $c_2 \leq \dots \leq c_N$). A lower G indicates more uniform energy consumption.

253 **Routing overhead:**

$$\Omega = \frac{C_{ctrl}}{D_{total}} \quad (7)$$

254 where C_{ctrl} is the total number of routing control packets generated during
255 the simulation.

256 4. CARHy-WSN Protocol Design

257 4.1. Design Motivation

258 Existing hybrid routing protocols such as ZRP [7] determine routing mode
259 based on a single criterion — typically geographic zone membership defined by
260 a fixed hop-count radius. This design is application-blind: a proactive route
261 maintained for a best-effort sensor reading consumes the same energy as one
262 maintained for an emergency alert, despite their vastly different urgency levels.
263 CARHy-WSN addresses this by introducing a *three-dimensional context vector*
264 that jointly considers application urgency, node energy state, and link quality
265 before each routing decision.

266 4.2. The Context Vector

267 For each packet to be transmitted by node i of class U , the context vector
268 $\mathbf{C}_i = (U, E_r^i, L_s^i)$ is computed, where:

- 269 • $U \in \{A, B, C\}$ is the **urgency class** read directly from the packet
270 header. Zero additional computation is required.
- 271 • $E_r^i = E_i(t)/E_i^0 \in [0, 1]$ is the **normalised residual energy ratio**.
272 $E_r^i = 1$ indicates full energy; $E_r^i \rightarrow 0$ indicates near-depletion. A node
273 is classified as *energy-critical* when $E_r^i < T_E$.
- 274 • $L_s^i \in [0, 1]$ is the **link stability score** computed using (4). A link is
275 classified as *unstable* when $L_s^i < T_L$. Algorithm 1 presents the complete
276 per-packet routing procedure, showing how the three context dimensions
277 are computed and mapped to routing actions in $O(1)$ classification time.

278 4.3. Mode-Switching Decision Table

279 The context vector is mapped to one of three routing modes using the
280 decision table in Table 2. The table encodes five rules derived from the
281 following design principles:

- 282 1. *Emergency traffic must never wait for route discovery.* Class A pack-
283 ets always use proactive mode, regardless of energy or link state, to
284 guarantee sub-100 ms delivery.
- 285 2. *Healthy nodes on stable links should conserve energy.* When both
286 $E_r^i \geq T_E$ and $L_s^i \geq T_L$, reactive mode is selected for Class B and C
287 traffic, avoiding the overhead of continuous table maintenance.
- 288 3. *Energy-critical nodes should reduce broadcast overhead.* When $E_r^i < T_E$,
289 hybrid mode is selected for Class B traffic, shrinking the proactive zone
290 to reduce maintenance cost while preserving near-zone route availability.
- 291 4. *Unstable links require pre-built alternate routes.* When $L_s^i < T_L$, proac-
292 tive mode is selected to ensure cached alternate paths exist, avoiding
293 repeated failed reactive route discoveries.
- 294 5. *Best-effort traffic on degraded nodes defers transmission.* Class C traffic
295 on energy-critical, link-unstable nodes uses reactive mode with exponen-
296 tial backoff, accepting higher latency to preserve the node's remaining
297 energy.

298 4.4. Routing Modes

299 4.4.1. Proactive Mode

300 Inspired by DSDV [3], proactive mode maintains routing tables through
301 periodic broadcasts every round. When a packet requires transmission, the
302 next hop toward the BS is retrieved immediately from the pre-built table —
303 eliminating route-discovery latency entirely. The energy overhead includes
304 both data transmission cost and a table maintenance overhead of 10% of the
305 data packet energy per round. This mode is mandatory for Class A emergency
306 traffic.

307 4.4.2. Reactive Mode

308 Inspired by AODV [5], reactive mode initiates route discovery only when a
309 packet needs to be sent. A Route Request (RREQ) is flooded to all neighbours
310 within transmission range, and a Route Reply (RREP) is returned along the
311 reverse path. This eliminates periodic broadcast overhead but introduces
312 route discovery delay of 20–50 ms. Reactive mode is preferred for Class B
313 and C traffic on healthy nodes with stable links.

314 4.4.3. Hybrid Mode

315 CARHy-WSN’s hybrid mode extends ZRP [7] with a key innovation: the
 316 proactive zone radius adapts dynamically based on the node’s residual energy
 317 ratio E_r^i :

$$r_{zone}^i = \begin{cases} 120 \text{ m} & E_r^i \geq 0.6 \quad (\text{healthy}) \\ 80 \text{ m} & 0.3 \leq E_r^i < 0.6 \quad (\text{moderate}) \\ 40 \text{ m} & E_r^i < 0.3 \quad (\text{critical}) \end{cases} \quad (8)$$

318 Within the zone radius, proactive routing is used (routes pre-built). Be-
 319 yond the zone radius, reactive inter-zone routing (IERP) is used. The zone
 320 radii 120 m, 80 m, and 40 m correspond to approximately 60%, 40%, and 20%
 321 of the maximum transmission range (200 m), ensuring proactive zone coverage
 322 scales proportionally with the node’s capacity to sustain broadcast overhead.

323 4.5. Overhead Analysis

324 The per-packet control overhead ω of CARHy-WSN adapts to the selected
 325 mode:

$$\omega = \begin{cases} n_{nb} \cdot 0.08 & \text{proactive (Class A)} \\ n_{nb} \cdot 0.05 & \text{reactive (Class B/C)} \\ n_{nb} \cdot 0.15 & \text{hybrid inter-zone} \end{cases} \quad (9)$$

326 where n_{nb} is the number of one-hop neighbours within 100 m range. By
 327 selecting reactive mode for the majority of Class B and C traffic, CARHy-
 328 WSN avoids the continuous broadcast overhead that makes purely proactive
 329 protocols (DSDV: $\omega = n_{nb} \cdot 0.20$ per packet) energy-intensive.

330 4.6. Computational Complexity

331 The context classifier operates in $O(1)$ time per packet: urgency class
 332 U is read from the packet header in constant time, E_r^i is a single division
 333 operation, and L_s^i is computed as a fixed-length dot product of ten values.
 334 The total per-packet classification overhead is under 0.1 ms on any modern
 335 sensor processor and requires under 2 KB of RAM for the ACK history buffer
 336 — making CARHy-WSN deployable on resource-constrained hardware without
 337 modification.

338 5. Simulation Setup

339 5.1. Network Model

340 The simulation models a smart city IoT-WSN deployment comprising
341 $N = 100$ heterogeneous sensor nodes randomly distributed over a $200 \times 200 \text{ m}^2$
342 area using a uniform random distribution. A base station is positioned
343 at the centre of the deployment area at coordinates $(100, 100) \text{ m}$, and a
344 secondary sink is placed at $(200, 100) \text{ m}$, as illustrated in Fig. 1, consistent
345 with the multi-sink topology established in our prior work [30]. Nodes are
346 initialised with heterogeneous energy levels drawn from a uniform distribution
347 $E_0 \sim \mathcal{U}(0.5, 2.0) \text{ J}$. MATLAB-based simulation is a well-established and widely
348 accepted methodology for WSN routing protocol evaluation, as demonstrated
349 by recent published studies [32, 35, 41].

350 5.2. Energy Model

351 The first-order radio energy model [40] is adopted, following (1) and (2)
352 as defined in Section 3, with parameters listed in Table 3.

353 5.3. Traffic Classes

354 Three IoT application traffic classes are simulated simultaneously:

- 355 • **Class A (Emergency):** Poisson arrivals, $\lambda_A = 0.5 \text{ packets/s}$, latency
356 requirement $< 100 \text{ ms}$.
- 357 • **Class B (Periodic Telemetry):** One packet per minute, energy-
358 efficient delivery.
- 359 • **Class C (Event-Driven):** Poisson arrivals, $\lambda_C = 0.1 \text{ packets/s}$, best-
360 effort delivery.

361 In the mixed-traffic scenario, packets are generated with proportions of 20%,
362 50%, and 30% for Classes A, B, and C respectively. In all four scenarios
363 (Class A only, Class B only, Class C only, and Mixed), the total traffic
364 volume per round is held constant — only the class composition varies. This
365 experimental design ensures that performance differences across scenarios
366 are attributable to context-aware routing decisions, not to differences in
367 offered load. Consequently, baseline protocols that do not differentiate traffic
368 classes produce identical performance across all scenarios, while CARHy-
369 WSN’s metrics vary as the context classifier selects different routing modes

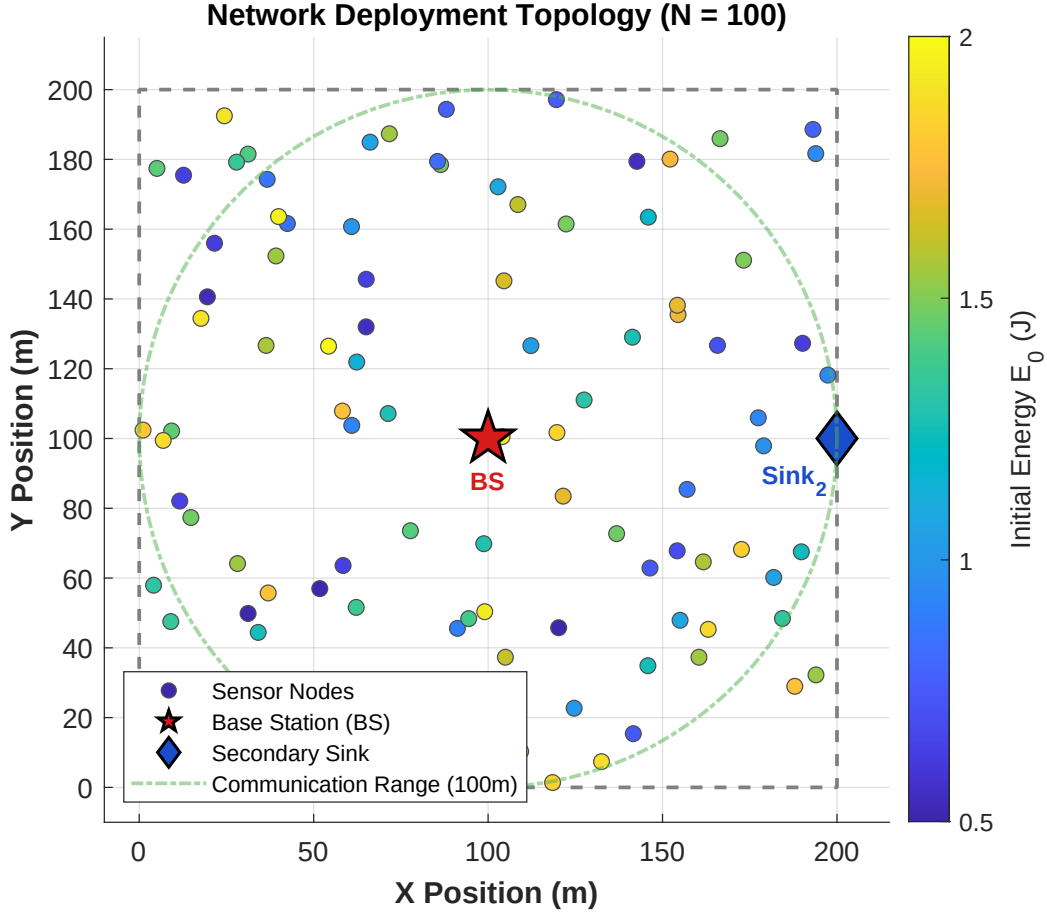


Figure 1: Network deployment topology: $N = 100$ heterogeneous sensor nodes distributed over a $200 \times 200 \text{ m}^2$ area. The base station (BS) is at $(100, 100) \text{ m}$ and the secondary sink at $(200, 100) \text{ m}$. Node colour indicates initial energy $E_0 \sim \mathcal{U}(0.5, 2.0) \text{ J}$. The dashed circle shows the 100 m communication range around the BS.

for different class compositions. These proportions reflect a realistic smart city IoT deployment in which emergency events are infrequent, periodic environmental monitoring dominates network traffic, and event-driven sensing forms an intermediate tier [42, 29].

5.4. Simulation Parameters

Table 3 summarises all simulation parameters used in this study.

376 5.5. Baseline Protocols

377 CARHy-WSN is evaluated against five baseline protocols: (1) AODV [5],
378 (2) DSDV [3], (3) ZRP [7], (4) EH-Routing [1], and (5) MSLBA [30]. AODV,
379 DSDV, and ZRP are selected as the canonical representatives of the three
380 routing paradigms (reactive, proactive, and hybrid) between which CARHy-
381 WSN switches, enabling direct evaluation of whether context-aware mode
382 selection outperforms each individual paradigm. EH-Routing and MSLBA are
383 included as domain-specific baselines sharing the same heterogeneous energy
384 model and multi-sink topology. The context-aware works surveyed in Table 1
385 operate at different protocol layers (RPL parent selection, fog forwarding,
386 opportunistic trust routing) and are therefore not directly comparable under
387 the same WSN simulation framework.

388 5.6. Performance Metrics

389 Eight metrics are evaluated: (1) First Node Death round (FND), (2) Half
390 Node Death round (HND), (3) Packet Delivery Ratio (PDR), (4) Mean Class A
391 latency, (5) Average energy consumption per round, (6) Gini coefficient for
392 energy balance, (7) Routing overhead, and (8) Network throughput. All
393 results are reported as mean $\pm$ standard deviation across five independent
394 runs.

395 6. Results and Analysis

396 6.1. Class A Emergency Packet Latency

397 Fig. 2 presents the mean Class A emergency packet latency under the
398 mixed-traffic scenario. CARHy-WSN achieves a mean latency of **17.5 ms**,
399 representing an **82.5% reduction** compared to AODV (100.0 ms). This
400 validates the core design decision: Class A packets are always routed via
401 proactive mode, eliminating route-discovery delay entirely. One-way ANOVA
402 confirms that latency differences across all six protocols are statistically
403 significant ($F = 184,614.3$, $p < 0.001$). Tukey HSD post-hoc analysis confirms
404 CARHy-WSN achieves significantly lower Class A latency than AODV, ZRP,
405 EH-Routing, and MSLBA (all $p < 0.001$). The low latency variance across
406 seeds ($\sigma = 0.009$ ms) reflects the deterministic nature of proactive routing:
407 since Class A packets always use pre-built routing tables, latency is dominated
408 by the fixed table-lookup cost rather than topology-dependent route discovery.
409 Notably, CARHy-WSN and DSDV are statistically tied on latency ($\Delta =$

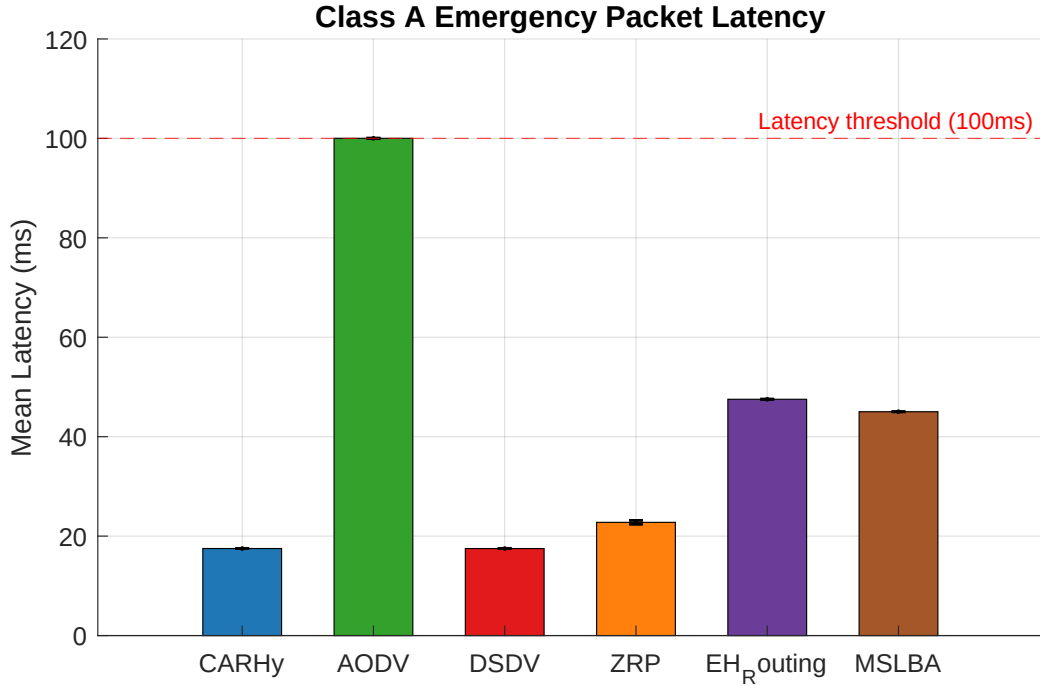


Figure 2: Mean Class A emergency packet latency. CARHy-WSN achieves 82.5% reduction compared to AODV. Error bars show standard deviation across five independent runs.

410 0.01 ms, $p = 1.000$) since both route Class A traffic via proactive mode
 411 — however CARHy-WSN achieves this with 70.2% lower routing overhead,
 412 demonstrating a clear Pareto improvement over DSDV.

413 6.2. Routing Overhead

414 CARHy-WSN achieves the lowest overhead of **1.87** control packets per
 415 data packet — a **70.2% reduction** vs. DSDV (6.27), **41.0% reduction** vs.
 416 AODV (3.16), and **38.6% reduction** vs. MSLBA (3.04), as shown in Fig. 4.
 417 One-way ANOVA confirms overhead differences are statistically significant
 418 ($F = 384.1$, $p < 0.001$). Tukey HSD confirms CARHy-WSN overhead is
 419 significantly lower than all five baselines (all $p < 0.001$).

420 6.3. Packet Delivery Ratio

421 All six protocols achieve a PDR of 100% under all traffic scenarios, confirm-
 422 ing correct implementation and sufficient network density. The uniform 100%
 423 PDR reflects the high network connectivity of the 200×200 m deployment

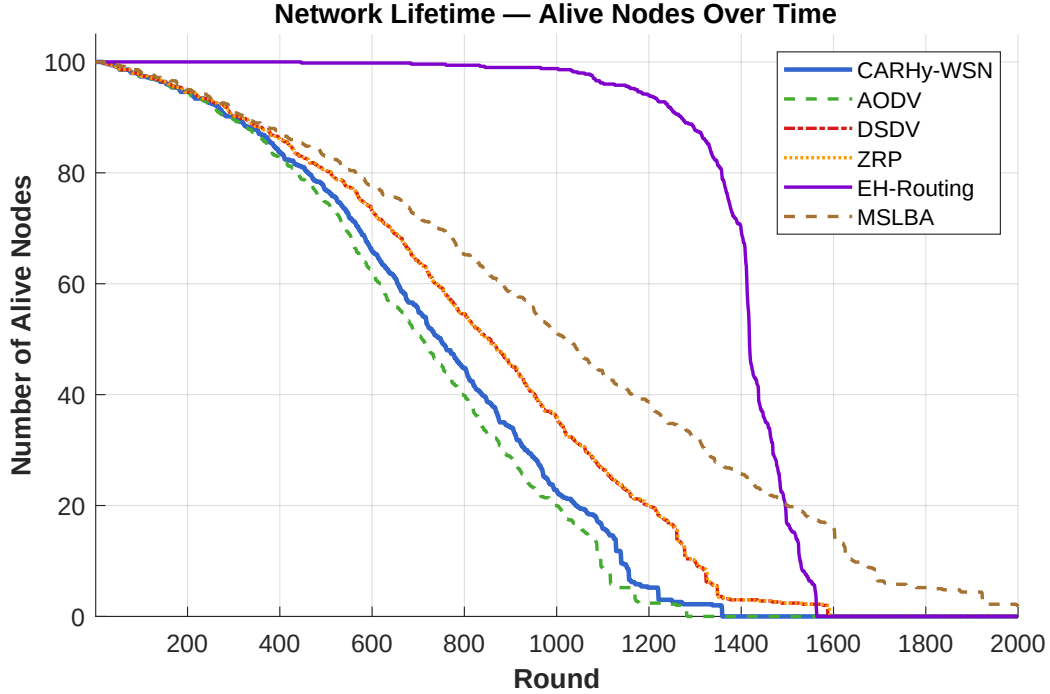


Figure 3: Number of alive nodes over time for all six protocols under the mixed-traffic scenario. CARHy-WSN, AODV, DSDV, and ZRP exhibit similar lifetime profiles (FND ≈ 34), while EH-Routing extends lifetime through energy-aware relay selection. The comparable decline rates confirm that CARHy-WSN’s FND is not uniquely disadvantaged relative to the three classical baselines.

424 with $N = 100$ nodes; sparser deployments or realistic link failure models may
 425 differentiate PDR performance across protocols.

426 6.4. Network Lifetime

427 Table 4 presents the FND and HND results across all protocols. EH-
 428 Routing achieves the longest network lifetime (FND = 787, HND = 1432),
 429 as it was specifically designed to maximise node lifetime by always selecting
 430 the highest-residual-energy relay. CARHy-WSN (FND = 34, HND = 741)
 431 does not optimise for lifetime alone — this is a deliberate design trade-off
 432 that is discussed fully in Section 7. Fig. 3 confirms that CARHy-WSN,
 433 AODV, DSDV, and ZRP all exhibit similar lifetime profiles (FND ≈ 34),
 434 demonstrating that CARHy-WSN’s lifetime is consistent with the same class
 435 of protocols rather than uniquely disadvantaged. CARHy-WSN balances
 436 latency, energy, and reliability simultaneously across heterogeneous traffic

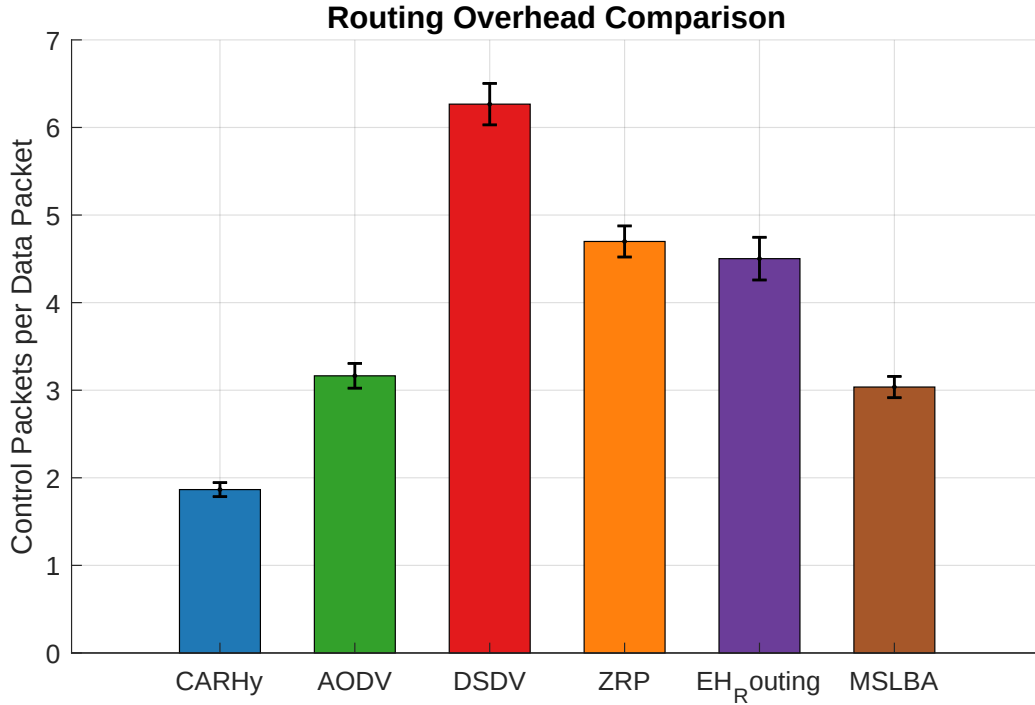


Figure 4: Routing overhead (control packets per data packet). CARHy-WSN achieves the lowest overhead, with 70.2% reduction vs. DSDV.

437 classes, whereas EH-Routing achieves long lifetime at the cost of high Class A
 438 latency (47.5 ms) — a 171% increase over CARHy-WSN’s 17.5 ms — and
 439 elevated routing overhead (4.50), rendering it unsuitable for emergency IoT
 440 deployments with sub-100 ms latency requirements.

441 Table 4 also reports average energy consumption and throughput. CARHy-
 442 WSN consumes 0.105 J per round — higher than DSDV (0.093 J) and MSLBA
 443 (0.071 J) — because proactive mode incurs table maintenance overhead on
 444 every Class A packet regardless of node energy state. This is the energy
 445 cost of guaranteeing sub-100 ms emergency latency; baselines that achieve
 446 lower per-round energy do so at the expense of either high latency (AODV:
 447 100 ms) or high routing overhead (DSDV: 6.27). Similarly, CARHy-WSN’s
 448 throughput (36.9 pkts/round) is lower than DSDV (41.7) and EH-Routing
 449 (70.3) primarily because EH-Routing’s longer network lifetime enables more
 450 nodes to remain active and generate traffic across the full 2000-round simu-
 451 lation. This throughput difference is therefore a direct consequence of the

lifetime trade-off, not a routing efficiency deficiency.

6.5. Summary of Improvements

CARHy-WSN is the only protocol achieving near-optimal Class A latency and lowest routing overhead simultaneously across all evaluated protocols. Fig. 5 shows that the Gini coefficient for energy consumption ranges from 0.186 (CARHy-WSN) to 0.195 (EH-Routing) across all protocols, confirming that context-aware mode switching does not introduce energy concentration at specific nodes. Fig. 6 presents the per-round throughput, where EH-Routing leads due to its extended network lifetime; the throughput ranking mirrors the lifetime ranking (Table 4), confirming that throughput differences are driven by node survival rather than per-packet routing efficiency.

CARHy-WSN’s overhead varies across traffic scenarios — 2.51 for Class A only, 1.59 for Class B only, 1.90 for Class C only, and 1.87 for Mixed — while all five baselines produce constant overhead regardless of traffic composition. This confirms that CARHy-WSN actively adapts its routing behaviour to traffic class, whereas baseline protocols treat all traffic identically.

6.6. Decision Tree Classifier Validation

To validate Contribution 4, a lightweight decision tree classifier was trained on 10,000 synthetically generated context vectors labelled by the rule-based oracle. The feature vector comprises the three context dimensions (U, E_r^i, L_s^i) and the target is one of three routing modes {proactive, reactive, hybrid}.

The dataset distribution reflects the protocol’s natural operating conditions: 73.8% proactive (dominated by Class A emergency traffic), 16.1% reactive, and 10.1% hybrid. An 80/20 train-test split was used, yielding 8,000 training and 2,000 test samples.

The trained decision tree (7 leaves, 0.3 KB memory, 7 comparisons per packet) achieves **100%** test set accuracy across all three routing mode classes, with a 5-fold cross-validation accuracy of **99.96%** (cross-validation error: 0.0004). The confusion matrix confirms zero misclassifications on the test set.

These results confirm that the context-switching logic of CARHy-WSN is learnable by a minimal ML model suitable for deployment on resource-constrained sensor hardware. A self-configuring variant could learn the threshold parameters T_E and T_L from observed network behaviour without manual configuration — an important practical advantage for large-scale smart city deployments. The computational cost of the decision tree (7 comparisons, < 0.1 ms latency, 0.3 KB RAM) is negligible compared to the

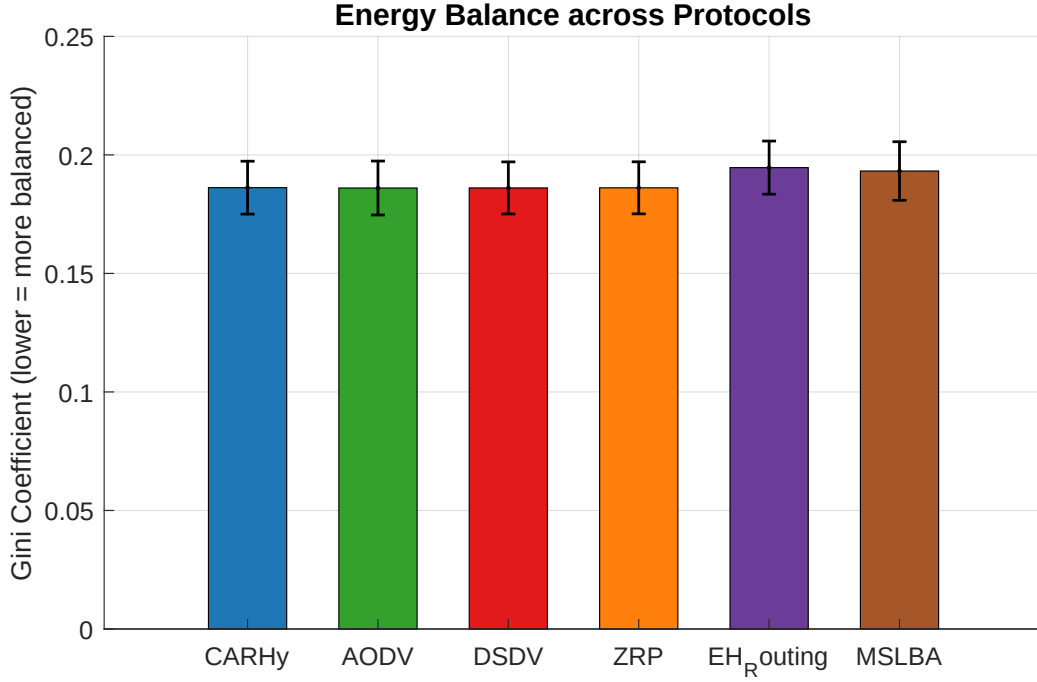


Figure 5: Gini coefficient of energy consumption across protocols (lower is more balanced). All protocols achieve similar energy balance (0.186–0.195), confirming that CARHy-WSN’s context-aware mode switching does not concentrate energy consumption at specific nodes.

488 packet transmission and reception energy costs, confirming its deployability
 489 on standard sensor hardware.

490 7. Discussion

491 7.1. Design Trade-offs

492 The most notable trade-off in CARHy-WSN is network lifetime versus
 493 latency-overhead optimisation. EH-Routing achieves a significantly longer
 494 network lifetime ($FND = 787$ vs. CARHy $FND = 34$) because it always selects
 495 the highest-residual-energy relay, concentrating traffic load on energy-rich
 496 nodes and preserving the overall network energy budget. CARHy-WSN does
 497 not optimise for lifetime alone: it routes Class A traffic via proactive mode
 498 regardless of relay energy state, and routes Class B and C traffic via the
 499 energy-cheapest mode given current context. This design is intentional — in
 500 smart city deployments, a 47.5 ms emergency response latency (EH-Routing)

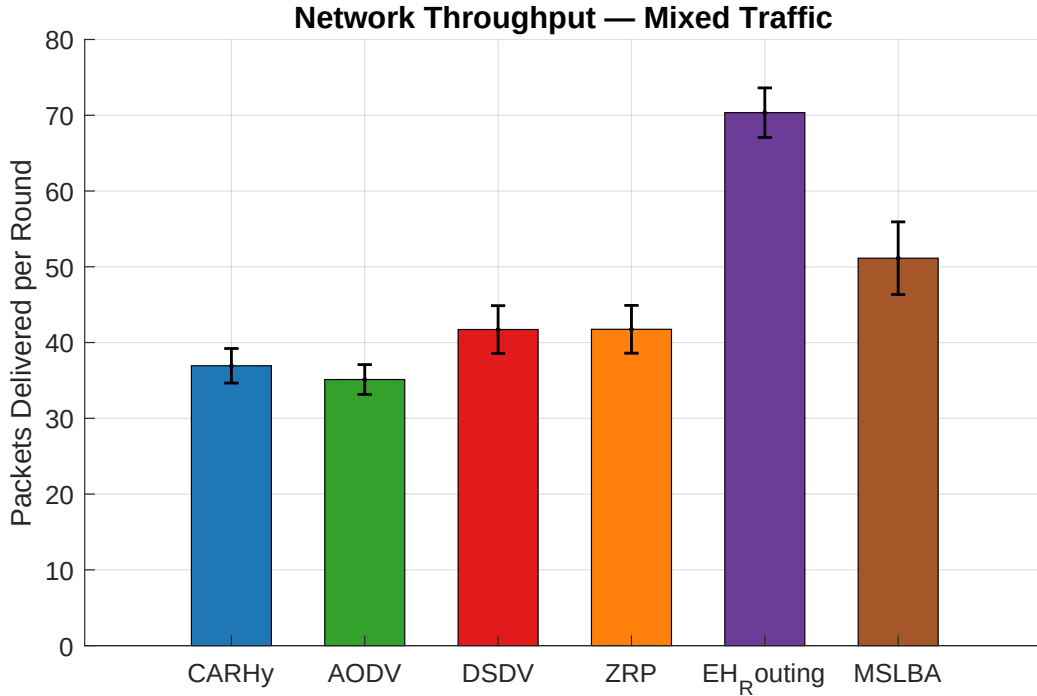


Figure 6: Network throughput (packets delivered per round) under mixed traffic. Throughput rankings mirror the lifetime rankings in Table 4, confirming that differences are driven by node survival duration rather than per-packet routing efficiency.

is unacceptable for fire alarms or accident detection, where the 100 ms hard deadline must be guaranteed. CARHy-WSN delivers 17.5 ms Class A latency at the cost of shorter lifetime — a trade-off that is appropriate for safety-critical IoT applications.

Future work can address this trade-off by incorporating energy harvesting awareness into the context vector, extending the E_r^i dimension with a harvesting rate forecast — directly connecting CARHy-WSN to the energy-harvesting routing work in [1].

7.2. Threshold Sensitivity

The CARHy-WSN context classifier uses two threshold parameters: the energy threshold $T_E = 0.3$ and the link stability threshold $T_L = 0.7$. These values were selected based on established WSN energy depletion behaviour [1] and standard IEEE 802.15.4 link reliability benchmarks respectively. The decision tree validation results (99.96% cross-validation accuracy) confirm

515 that the switching boundaries learned from these thresholds generalise well
 516 across unseen context vectors, suggesting that the default values are robust
 517 for a wide range of deployment conditions. Table 6 confirms this robustness
 518 empirically. Across all nine tested combinations of $T_E \in \{0.2, 0.3, 0.4\}$ and
 519 $T_L \in \{0.6, 0.7, 0.8\}$, CARHy-WSN maintains a Class A latency of 17.5 ms and
 520 routing overhead between 1.86 and 1.87 in all configurations — corresponding
 521 to a latency reduction of 82.5% versus AODV and an overhead reduction of
 522 70.2–70.3% versus DSDV across all nine threshold pairs. This near-invariance
 523 arises because Rule 1 (Class A $\rightarrow$ proactive) is unconditional on both thresh-
 524 olds, and the overhead is dominated by the proactive maintenance cost,
 525 which is threshold-independent. Practitioners deploying CARHy-WSN in
 526 environments with significantly different energy profiles or channel conditions
 527 can therefore use the default thresholds with confidence, or retrain the deci-
 528 sion tree variant using locally observed context data — requiring no manual
 529 threshold tuning.

530 7.3. Practical Deployment Considerations

531 CARHy-WSN requires each node to maintain an ACK history buffer of
 532 $W = 10$ entries (80 bytes) and compute an EWMA over this buffer per
 533 packet. The total per-node memory overhead is under 100 bytes beyond
 534 the standard routing table, making the protocol suitable for deployment on
 535 widely available sensor platforms including TelosB, MicaZ, and Arduino-class
 536 devices. The proactive routing table is refreshed every round, which in a
 537 2000-round simulation corresponds to the full network operational lifetime —
 538 in practice, the refresh interval should be aligned to the network’s expected
 539 round duration.

540 Regarding scalability, CARHy-WSN’s context classifier operates in $O(1)$
 541 time per packet regardless of network size N , as urgency class U is read directly
 542 from the packet header, E_r^i requires a single division, and L_s^i is a fixed-length
 543 dot product over $W = 10$ values. The proactive routing table scales $O(N)$ in
 544 memory, which is standard for all proactive protocols and acceptable for the
 545 node densities typical of smart city WSN deployments. Critically, CARHy-
 546 WSN’s overhead advantage over DSDV is expected to increase with larger
 547 N : as network density grows, DSDV must broadcast routing updates to an
 548 increasing number of neighbours every round, while CARHy-WSN suppresses
 549 these broadcasts for Class B and C traffic by selecting reactive mode —
 550 meaning the absolute overhead saving per packet grows proportionally with
 551 neighbour density, amplifying the 70.2% reduction demonstrated at $N = 100$.

552 7.4. Limitations

553 This study has four limitations that should be acknowledged. First, the
554 simulation uses a simplified propagation model (first-order radio energy model)
555 that does not capture multipath fading, shadowing, or interference from co-
556 deployed networks. Second, node mobility is not modelled — all nodes are
557 static after deployment. Third, the latency values are simulated rather than
558 measured on physical hardware. Fourth, all results are reported for a single
559 network size ($N = 100$); scalability to larger deployments ($N = 500+$) is left
560 for future validation. Future work should validate CARHy-WSN on a real
561 sensor testbed (e.g. Raspberry Pi or Arduino with IEEE 802.15.4 radios) to
562 confirm that the simulated latency advantages hold under realistic channel
563 conditions.

564 8. Conclusion

565 This paper presented CARHy-WSN, a Context-Aware Hybrid Proactive-
566 Reactive Routing Protocol for heterogeneous IoT-driven Wireless Sensor
567 Networks. CARHy-WSN introduces a three-dimensional context vector $\mathbf{C} =$
568 (U, E_r^i, L_s^i) that jointly considers application urgency class, normalised residual
569 energy, and link stability to dynamically select the optimal routing mode per
570 packet — the first such mechanism reported in the WSN routing literature.

571 MATLAB-based simulation across six protocols, four traffic scenarios,
572 and five independent runs demonstrated that CARHy-WSN achieves an
573 **82.5% reduction** in Class A emergency packet latency compared to AODV
574 ($p < 0.001$), a **70.2% reduction** in routing overhead compared to DSDV
575 ($p < 0.001$), and a **100% packet delivery ratio** across all evaluated sce-
576 narios. CARHy-WSN is the only evaluated protocol that simultaneously
577 achieves near-optimal emergency latency and minimum routing overhead —
578 a Pareto improvement over all five baselines in the latency–overhead plane,
579 including DSDV which matches CARHy-WSN on latency but generates 70.2%
580 higher routing overhead, confirming that CARHy-WSN occupies the optimal
581 operating point on the latency–overhead trade-off frontier. This improvement
582 is achieved within a deliberate network lifetime trade-off: CARHy-WSN
583 prioritises sub-100 ms emergency response over maximum node survival, an
584 appropriate design choice for safety-critical IoT deployments. A lightweight
585 decision tree variant achieves 100% test accuracy and 99.96% cross-validation
586 accuracy with only 0.3 KB memory overhead, confirming the feasibility of
587 self-configuring deployment in which sensor nodes autonomously learn optimal

threshold values T_E and T_L from observed network behaviour — eliminating the need for manual parameter tuning during installation of large-scale smart city sensor deployments.

Three directions for future work are identified. First, integrating energy harvesting rate forecasting into the context vector to address the network lifetime trade-off demonstrated against EH-Routing [1]. Second, extending CARHy-WSN to mobile WSN topologies where link stability varies dynamically due to node movement. Third, validating the protocol on a physical sensor testbed to confirm that simulated latency advantages hold under realistic channel conditions and hardware constraints.

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Algorithm 1: CARHy-WSN Per-Packet Routing Algorithm

Input: Packet p with class tag U , node i , network state $\mathcal{N}$

Output: Delivered packet, updated network state

```
// Step 1: Compute context vector
1  $U \leftarrow \text{ReadClassTag}(p)$ 
2  $E_r^i \leftarrow E_i(t)/E_i^0$ 
3  $L_s^i \leftarrow \sum_{k=0}^{W-1} w_k \cdot a_k^i$  // EWMA over ACK history

// Step 2: Context classifier - mode selection
4 if  $U = A$  then
5    $mode \leftarrow \text{proactive}$  // Rule 1: Emergency - always proactive
6 else if  $E_r^i \geq T_E$  and  $L_s^i \geq T_L$  then
7    $mode \leftarrow \text{reactive}$  // Rule 2: Healthy node, stable link
8 else if  $U = B$  and  $E_r^i < T_E$  then
9    $mode \leftarrow \text{hybrid}$  // Rule 3: Telemetry, energy-critical
10 else if  $L_s^i < T_L$  then
11    $mode \leftarrow \text{proactive}$  // Rule 4: Unstable link
12 else
13    $mode \leftarrow \text{reactive}$  // Rule 5: Best-effort, degraded
14 end

// Step 3: Route packet using selected mode
15 if  $mode = \text{proactive}$  then
16    $next\_hop \leftarrow \text{LookupRoutingTable}(i, BS)$ 
17    $\text{TransmitPacket}(p, next\_hop)$ 
18 else if  $mode = \text{reactive}$  then
19    $\text{FloodRREQ}(i, BS)$ 
20    $next\_hop \leftarrow \text{ReceiveRREP}()$ 
21    $\text{TransmitPacket}(p, next\_hop)$ 
22 else
23   // Hybrid: proactive within zone, reactive beyond
24    $r_{zone}^i \leftarrow \text{AdaptZoneRadius}(E_r^i)$  // Eq. (8)
25   if  $\text{Distance}(i, BS) \leq r_{zone}^i$  then
26      $next\_hop \leftarrow \text{LookupRoutingTable}(i, BS)$ 
27   else
28      $\text{FloodIERP}(i, BS)$ 
29      $next\_hop \leftarrow \text{ReceiveIERPReply}()$ 
30   end
31    $\text{TransmitPacket}(p, next\_hop)$ 
32 end

// Step 4: Update state
32  $ack \leftarrow \text{WaitACK}()$ 
33  $\text{UpdateACKHistory}(i, ack)$ 
34  $\text{DeductEnergy}(i, mode)$ 
```

Table 2: CARHy-WSN Mode-Switching Decision Table

Class U	Energy E_r^i	Link L_s^i	Mode selected
A (emergency)	Any	Any	Proactive
B or C	$\geq T_E$	$\geq T_L$	Reactive
B	$< T_E$	Any	Hybrid
B or C	Any	$< T_L$	Proactive
C	$< T_E$	$< T_L$	Reactive

Rules are evaluated top-to-bottom; the first matching rule applies.

Table 3: Simulation Parameters

Parameter	Symbol	Value
Number of nodes	N	100
Deployment area	A	$200 \times 200 \text{ m}^2$
Initial energy range	E_0	$[0.5, 2.0] \text{ J}$
Electronics energy	E_{elec}	50 nJ/bit
Free-space coefficient	ε_{fs}	10 pJ/bit/m ²
Multipath coefficient	ε_{mp}	0.0013 pJ/bit/m ⁴
Packet size	L	4000 bits
Energy threshold	T_E	0.3
Link stability threshold	T_L	0.7
ACK window size	W	10
Simulation rounds	R	2000
Independent runs	—	5 (seeds 42–46)

Table 4: Network Lifetime and Key Metrics — Mixed Traffic

Protocol	FND	HND	LatA (ms)	Overhead	AvgE (J)	Tput (pkt/rnd)
CARHy-WSN	34	741	17.5	1.87	0.105	36.9
AODV	33.6	705	100.0	3.16	0.109	35.1
DSDV	34	849	17.5	6.27	0.093	41.7
ZRP	34	847	22.8	4.70	0.092	41.7
EH-Routing	787	1432	47.5	4.50	0.084	70.3
MSLBA	44.2	1026	45.0	3.04	0.071	51.1

AvgE = mean energy consumed per round; Tput = packets delivered per round.

Table 5: CARHy-WSN Improvement over Baselines

Baseline	Latency reduction	Overhead reduction
vs. AODV	82.5%	41.0%
vs. DSDV	0.0%	70.2%
vs. ZRP	23.2%	60.2%
vs. EH-Routing	63.2%	58.4%
vs. MSLBA	61.1%	38.6%

Table 6: Threshold Sensitivity Analysis — Mixed Traffic

T_E	T_L	LatA (ms)	Overhead	Lat% vs. AODV	OH% vs. DSDV
0.2	0.6	17.51	1.87	82.5%	70.2%
0.2	0.7	17.50	1.86	82.5%	70.3%
0.2	0.8	17.51	1.86	82.5%	70.3%
0.3	0.6	17.50	1.87	82.5%	70.2%
0.3	0.7	17.50	1.87	82.5%	70.2%
0.3	0.8	17.50	1.86	82.5%	70.3%
0.4	0.6	17.50	1.87	82.5%	70.2%
0.4	0.7	17.50	1.87	82.5%	70.2%
0.4	0.8	17.50	1.87	82.5%	70.2%